

On Hurwitz Quaternions and Lagrange's Four-Square Theorem

Erin Compaan and Cynthia Wu
SPWM 2011

Goal

- ▶ Lagrange's Four Square Theorem:

Any natural number N can be represented as

$$N = a^2 + b^2 + c^2 + d^2$$
 where a, b, c , and d are integers.

Our goal is to prove this theorem using Hurwitz Quaternions.

History of the Problem

- ▶ Diophantus – ca. 200 A.D.
- ▶ Bachet – 1621
- ▶ Fermat – 17th c.
- ▶ Lagrange – 1770



Quick Review of Quaternions

- ▶ Denoted by $\mathbb{H}$
- ▶ Members of a non-commutative division algebra
- ▶ Form of quaternions:

$$a + bi + cj + dk$$
 where a, b, c , and d are real numbers
- ▶ Fundamental formula of quaternion algebra:

$$i^2 = j^2 = k^2 = ijk = -1$$

Hurwitz Quaternion

- ▶ Not that much of a difference!

$$H = \{a + bi + cj + dk \in \mathbb{H} : a, b, c, d \in \mathbb{Z} \text{ or } a, b, c, d \in \mathbb{Z} + \frac{1}{2}\}$$

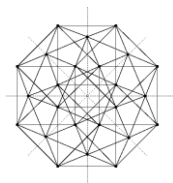
- ▶ So now a, b, c , and d are either *all* integers or *all* half integers
- ▶ Half integers: all numbers that are half of an odd integer - the set $\mathbb{Z} + \frac{1}{2}$.
 - EG: $7/2$, $-13/2$, 8.5

Hurwitz Primes

- ▶ A Hurwitz quaternion α is *prime* if it is divisible only by the quaternions $\pm 1, \pm i, \pm j, \pm k$, and $\pm \frac{1}{2} \pm \frac{1}{2}i \pm \frac{1}{2}j \pm \frac{1}{2}k$, and multiples of α with these.
- ▶ A Hurwitz quaternion β *divides* α if there exists a Hurwitz quaternion φ such that $\alpha = \beta\varphi$ or $\alpha = \varphi\beta$.

Lipschitz Quaternions

- ▶ A *Lipschitz quaternion* is a quaternion of the form $a + bi + cj + dk$, with $a, b, c, d \in \mathbb{Z}$.
 - E.g. $1 + 7i - 83j + 12k$.



Four Square Theorem

Any natural number N can be represented as

$$N = a^2 + b^2 + c^2 + d^2$$
 where a, b, c , and d are integers.

Prove this using Hurwitz Quaternions:

$$H = \{a + bi + cj + dk \in \mathbb{H} : a, b, c, d \in \mathbb{Z} \text{ or } a, b, c, d \in \mathbb{Z} + \frac{1}{2}\}$$

Lemmas and Preliminaries

- ▶ If p is a prime $p = 2n + 1$, $n \in \mathbb{N}$, then there are $l, m \in \mathbb{Z}$ such that p divides $1 + l^2 + m^2$.
- ▶ If a Hurwitz prime divides a product of Hurwitz quaternions $\alpha\beta$, then the prime divides α or β .

Lemma: Four Square Identity

- ▶ If two numbers can be written as a sum of four integer squares, then so can their product.
- ▶ Proof: Suppose that $u = a^2 + b^2 + c^2 + d^2$ and $v = w^2 + x^2 + y^2 + z^2$.

Then $u = \|a + bi + cj + dk\|^2 = \|\alpha\|^2$ and
 $v = \|w + xi + yj + zk\|^2 = \|\beta\|^2$, for some Lipschitz quaternions α and β .

Then

$$uv = (\|\alpha\| \|\beta\|)^2 = \|\alpha\beta\|^2 = \|A + Bi + Cj + Dk\|^2 = A^2 + B^2 + C^2 + D^2 \text{ for some } A, B, C, D \in \mathbb{Z}.$$

Proof

- ▶ Base Cases

$$1 = 1^2 + 0^2 + 0^2 + 0^2$$

$$2 = 1^2 + 1^2 + 0^2 + 0^2$$

Proof for Non-Hurwitz Primes

- ▶ Suppose p is an odd prime which has a non-trivial Hurwitz factorization $p = (a + bi + cj + dk)\alpha$.
- ▶ Conjugating: $p = \bar{p} = \bar{\alpha}(a - bi - cj - dk)$.
- ▶ Multiplying the equations:
 $p^2 = (a + bi + cj + dk)\alpha\bar{\alpha}(a - bi - cj - dk)$
 $= (a^2 + b^2 + c^2 + d^2)\|\alpha\|^2$
- ▶ Since p is prime, the factors of p^2 must both be p . Thus $p = a^2 + b^2 + c^2 + d^2$.
- ▶ If a, b, c , and d are integers, we're done.
- ▶ If not, we can still show that p is a sum of four integer squares.

Proof for a General Case

- ▶ Now let p be an odd prime. Then there exist integers l and m such that p divides $1 + l^2 + m^2$.
- ▶ Then p divides $(1 + li + mj)(1 - li - mj)$. By the previously stated lemma, if p were a Hurwitz prime, it must divide one of these factors.
- ▶ But this would imply that $\frac{1}{p} + \frac{l}{p}i + \frac{m}{p}j$ or $\frac{1}{p} - \frac{l}{p}i - \frac{m}{p}j$ is a Hurwitz integer, a contradiction.
- ▶ Thus p is not a Hurwitz prime.

Proof for a General Case

- ▶ Since p is not a Hurwitz prime, we can apply our previous conclusion and say that p is a sum of four integer squares.
- ▶ We now have that 1, 2, and all odd primes can be written as a sum of four squares.
- ▶ By the Four Squares identity, every natural number can be written as a sum of four squares.

Related Two-Square Theorem

- ▶ Fermat's Two Square Theorem: If a prime p is of the form $4n + 1$ for some $n \in \mathbb{N}$, then $p = a^2 + b^2$ for some $a, b \in \mathbb{Z}$.
- ▶ *Gaussian integers*: Complex numbers with integer coefficients.
- ▶ *Gaussian integer prime*: A Gaussian integer z which is divisible only by ± 1 or $\pm i$, or products of z with these.
- ▶ Lemma: For any prime p of the form $4n + 1$, $n \in \mathbb{N}$, there exists an integer m such that p divides $1 + m^2$.
- ▶ Lemma: If a Gaussian integer prime p divides $\alpha\beta$ for some Gaussian integers α and β , then p divides α or p divides β .

Proof of Fermat's Two Square Theorem

- ▶ Suppose p is a prime of the form $4n + 1$ for some $n \in \mathbb{N}$.
- ▶ Then p divides $1 + m^2 = (1 + mi)(1 - mi)$ for some $m \in \mathbb{N}$.
- ▶ Since p divides neither factor of $1 + m^2$, p is not a Gaussian prime.
- ▶ Then p has a nontrivial factorization in the Gaussian integers $p = (a + bi)(x + yi)$.

► Conjugating and multiplying equations

- $$p^2 = (a + bi)(x + yi)\overline{(a + bi)(x + yi)}$$
- $$= (a + bi)(a - bi)(x + yi)(x - yi)$$
- $$= (a^2 + b^2)(x^2 + y^2).$$
- Since p is prime and the factorization was nontrivial, the factors $a^2 + b^2$ and $x^2 + y^2$ are equal to p .
- Thus p can be written as a sum of two integer squares.

Lagrange



- Various contributions
- Calculus of Variations, Lagrange Multipliers, PDE's
- Prolific writer
- Proved four square theorem in 1770
- Meticulous and shy

Joseph-Louis Lagrange

Hurwitz



- Born to a Jewish family
- Number theorist
- Mostly contributed to number theory and algebras
- Sickly

Adolf Hurwitz

Lipschitz



- German mathematician
- Investigated number theory, Bessel functions, PDE's
- Work on quadratic mechanics influenced Einstein
- Also very sickly

Rudolf Lipschitz

Concluding Remarks

Any natural number N can
be represented as

$$N = a^2 + b^2 + c^2 + d^2$$

where a , b , c , and d are
integers

► Hurwitz quaternions can
be useful in a variety of
ways!

